Internship Report
Backend Development at AO ForteBank

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Program: 6B06101 Computer Science

Period: March 9, 2026 – May 2, 2026

Introduction

The internship goal was to gain hands-on experience in backend development within a corporate IT environment. Primary focus areas: designing and implementing RESTful APIs, working with relational databases, and creating small automation tools. The placement provided exposure to production workflows, code review practices, and cross-team coordination.

About AO ForteBank

Context

AO ForteBank operates in banking and financial services. The IT department supports development, operations, and internal support for systems including CRM, transaction processing, and employee-facing tools.

- Industry: Banking & Financial Services
- Department: IT — Dev / Support / Ops
- Key systems: CRM, transaction processing, internal dashboards

Role & Responsibilities

Role: Junior Backend Developer (Intern)

Main tasks:

- Implement REST APIs
- Design lightweight database schemas (SQLite)
- Write integration scripts

Tech stack:

Python, SQLite | Flask | Java (core) | Git | Telegram Bot API

Bank Client System (Python + SQLite)

Features:

- Add / view / delete clients
- Update account balances
- Transaction history per client

✅ Week 3 (March 23–29)

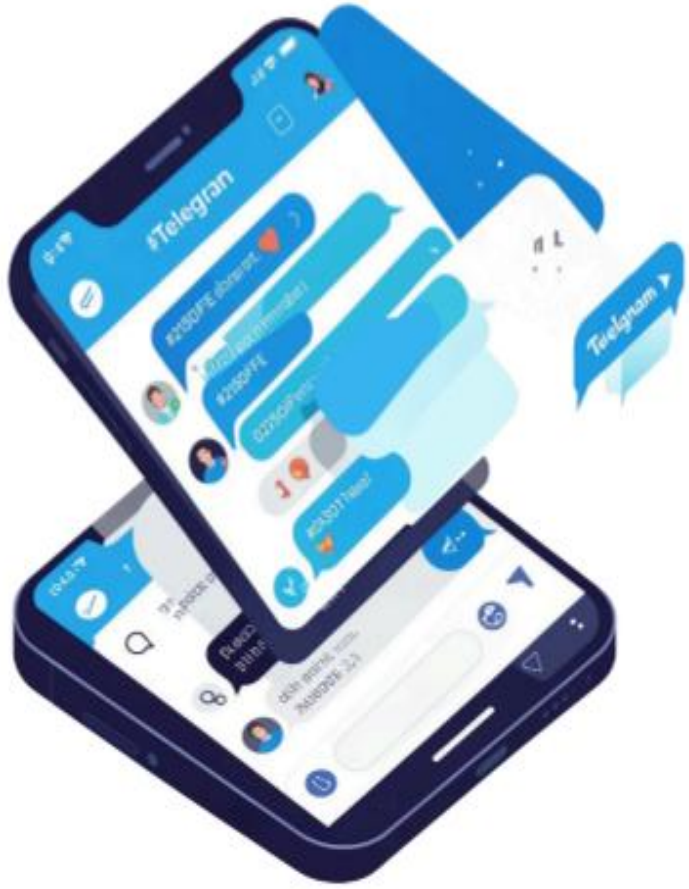
Expense Tracker Web App (Flask + SQLite)

Features:

- Add / view expenses via web forms
- Server-side total calculation
- Basic filtering
- Responsive UI

✅ Week 4 (March 30 – April 5)

Mini Chatbot (Python + SQLite + API)



Capabilities:

- Answers frequently asked questions (FAQ)
- Stores conversation logs in SQLite
- REST API for user interaction

✅ Week 5 (April 6–12)

Java Expense Tracker (Console-based)

Features:

- Add expenses
- Show all expenses
- Calculate total spending

✅ Week 6 (April 13–19)

Challenges & Solutions

SQLite Database Locking

Issue: occasional "database is locked" errors under concurrent writes. Mitigation: implemented short-lived connections, retries with exponential backoff, and moved heavier operations to background tasks.

Bot State Management

Issue: chat state desynchronization when users interrupt flows. Mitigation: explicit state machine per user, idempotent handlers, and clearer prompts with cancel/confirm options.

Input Validation

Issue: malformed inputs led to exceptions. Mitigation: strict server-side validation, sanitized parsing of amounts, and user-friendly error messages.

What I Learned & Next Steps

Key Takeaways

- Backend architecture: designing clear, RESTful endpoints and modular route structure.
- Database design: simple normalized schemas, transactions, and concurrency considerations with SQLite.
- Practical debugging: diagnosing concurrency and state issues in real systems.
- Software practices: use of Git, code reviews, and incremental development.

Recommended Next Steps

To continue maturing the projects:

1. Move persistent storage to a client-server RDBMS for concurrent access (e.g., PostgreSQL).
2. Introduce unit and integration tests for API endpoints and bot handlers.
3. Extend the web UI for categorization, reporting, and CSV export.
4. Document architecture and hand off guidelines for maintainers.

Thank you to AO ForteBank mentors and supervisors for guidance and code reviews during the internship.

Thank you